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Maximizing Auditory Cortex Output: Optimal Source and Detector Placement on the Scalp

Final Report

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Abstract

Functional Near Infrared Spectroscopy (fNIRS) is becoming popular day by day in the field of hearing research. This is because of it's robustness and cost effectiveness. Proper placement of fNIRS sources and detectors on the scalp is crucial for capturing the best information when the auditory cortex is stimulated. NIRFAST, an imaging technique, is commonly used to reconstruct images from functional data that has been collected. In this research, I aim to determine the optimal placement of fNIRS sources and detectors when the auditory cortex is stimulated, using the NIRFAST modeling package for simulations. When the human brain detects sound, the auditory cortex is activated, leading to changes in blood flow within the surrounding brain tissue. NIRFAST can simulate these changes in blood flow, allowing us to replicate this process for analysis. Through this process, I aim to identify the optimal arrangement of source-detector (SD) pairs to ensure the simulated setup captures the maximum information from the auditory cortex. These configurations can later be applied in practical situations. The primary goal of this research is to determine the best positions for detecting auditory cortex stimulation, regardless of sound duration or type.

Acknowledgements

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CHAPTER 1

Introduction

Human brain has two auditory cortices one on each hemisphere. Both are located about 15 millimeters above the ears and 10 millimeters depth from the scalp. The left auditory cortex primarily processes speech and voice sounds, while the right one is more involved with music and non-verbal auditory signals. This is an important aspect to understand how brain process complex auditory information and how different type of sound with having different wavelength and amplitude such as language and music represent to the brain. To understand how sound works in the brain specifically in the auditory cortices researchers often use neuroimaging techniques like Funtional Near Infrared Spectroscopy Peng *et al.*[13]. Now a days Functional Near-Infrared Spectroscopy (fNIRS) is becoming increasingly popular for it's non-invasive nature, cost effectiveness and most importantly it can measure brain activity which is associated with sound.

fNIRS is a very safe and effective method. It uses near infrared light to measure brain activity. This process starts with placing light sources in the human scalp. These light sources emit near infrared light. These injected ligth propagates through the scalp, skull to the brain tissues. During this propagation near infared light interacts with substances like hemoglobin,the primary carrier of oxygen in the blood. By these substances in the brain tissue it partly gets absorbed or scattered. Rest of the near infrared light comes back to the detectors after the travel. By detecting the light by the detectors that comes back, fNIRS can measure the levels of oxygenated and deoxygenated hemoglobin in the blood Chen *et al.* [1]. These measurements allow researchers to track changes in brain activity based on blood oxygen levels for a region of Interest(ROI).

One of the reasons fNIRS is so valuable is that it's safe, non-invasive, and comfortable for participants. This makes it ideal for studying brain function in various environments. When someone hears a sound, the auditory cortex in their brain responds by increasing blood flow, which changes the levels of

oxygenated and deoxygenated hemoglobin. fNIRS can capture these changes in the level of oxygenated and deoxygenated hemoglobin, offering a non-invasive method to monitor real time brain activity. This ability to map and track the brain activity using near infrared light makes it a valuable tool for neuroimaging, clinical diagnosis and development of auditory devices.

fNIRS has lot of advantages such as safety, comfortness for the participants etc. There is an other neuroimaging technique called fMRI where subject has to be in a large, noisy scanner for the research purpose. Whereas fNIRS's equipment is compact, safe and offer research in natural environment. This technique can be so portable that researchers can measure brain activity in schools, hospitals even in homes. Moreover, this technique is safe for various type of population including infants or elderly persons.

Despite having so much advantages of fNIRS to get the maximum level of information or accurate reading from auditory cortex, the positioning of its sources and detectors on the scalp is very important. The quility of fNIRS data is highly dependent on the arrangment of the optodes(sources and detector) on the scalp. The arrangement of these sources and detectors directly impacts how well fNIRS can capture brain activity for a specific brain region, and they need to be placed around 15 millimeters above the ears as we know the position of auditory cortex but it is hard to find the exact position of the optodes that lead to best quility of data. A suboptimal formation of the optodes can lead to a poor quality of data. Therefore researchers need to find out the optimal position sources and detectors so that these optodes can capture important information when auditory cortex stimulates.

Although we have a general understanding of where the auditory cortex is located on the scalp, the exact positioning of fNIRS sources and detectors for optimal results is still unclear. This is particularly important given that the brain's curvature and the complexity of light propagation through biological tissues can affect how well fNIRS captures concentration changes of HbO and HbR in the auditory cortex. This is where my research comes in. Using NIRFAST Dehghani *et al.* [4], an image reconstruction modeling package in MATLAB, I aim to find the best positions and arrangements of the fNIRS sources and detectors through simulations. This research allows me to create a realistic situation about how auditory cortex responds to a stimuli and how fNIRS propagates through brain tissue to capture brain activity.

In this simulation I use 'headmesh' model which is acting as a human head with structural and optical properties of real human head to simulates practical situation when fNIRS works. The 'headmesh' represents realistic model of brain tissue including scalp, skull cerebral cortex. In this simulated environment, I create an activation, or "blob" in the auditory cortex to imitate neural activity to the sound. This 'blob' is the changes of HbO and HbR in real situation resopond to the sound in the auditory cortex. Then, I place optodes (fNIRS sources and detectors) on the headmesh to capture this simulated brain activity.

The core part of my research is identify the best optodes position to get the best information from the auditory cortex when it stimulates. By systematically

varying the positions of the sources and detectors, I can assess how different arrangements affect the fNIRS signals and identify the channels that carry the most significant information. In this process I figure out how much absorption or scattering occurs for each channel due to the activation in the auditory cortex or 'blob'. My goal is to find out the best optodes position that maximise the each channels data capture quality.

After identifying these optimal placements, I plan to conduct practical tests on individuals to verify the effectiveness of the configurations. The ultimate goal of this project is to provide clear guidelines for positioning fNIRS equipment to improve the accuracy of measuring auditory cortex activity.

Moreover, researching how the auditory cortex activates can have a significant impact in various areas. It can lead us to a better understanding of hearing impairments and auditory processing disorders. By obtaining precise fNIRS measurements, we could improve the development of hearing devices and cochlear implants, ultimately helping people with hearing difficulties.

CHAPTER 2

Related Work and Background

2.1 Background

In this section I talk about the important things and terms that I use in my research work.

2.1.1 Auditory Cortex

Auditory cortex is a region in brain which is responsible for processing sound that human detects. It is located in the superior temporal gyrus of the temporal lobe and is traditionally divided into primary and secondary auditory areas. The primary auditory cortex is responsible process sounds with different frequencies Moerel *et al.* [10]. Auditory cortex not only work with basic sound but also responsible for complex auditory functions such as speech perception, sound localization, and auditory memory Hackett *et al.* [7]. There are two auditory cortices in a normal human brain. They are 15 mm above each ear and 10 mm depth from the scalp. The left auditory cortex is responsible for processing human speech and the right one process music and other kind of sounds.

2.1.2 Oxygenated and Deoxygenated Hemoglobin in Auditory Cortex

For processing sound auditory cortex needs energy and oxygen. When a neural activity happen specifically when a person hear something the concentration of Oxygenated Hemoglobin(HbO) and Deoxygenated(HbR) Hemoglobin changes. When neurons in the auditory cortex become active, there is an increase in metabolic demand, leading to a process called the hemodynamic response. In this case it becomes a rush situation of oxygenated blood in auditory cortex. It

cause rise in HbO and a decrease in HbR Obrig *et al.*[11]. This change is helping the study of cortical activities to modalities like Functional Near-Infrared Spectroscopy(fNIRS) and Functional Magnetic Resonance Imaging(fMRI). For example, during the stimulation in auditory cortex(hearing speech, music) the HbO rises in that specific region Zatorre *et al.*[16]. So, these two substances in the brain helps researchers in many way to study more about cortical activity as well as other regions in the brain.

2.1.3 Functional Near Infrared Spectroscopy(fNIRS)

Functional Near Infrared Spectroscopy is an optical imaging technique which is non-invasive, cost effective. It is use to monitor brain activity by measuring the changes in some substances in blood specifically HbO and HbR. It emittes near infrared light (650nm-900nm) into the scalp then the light penetrates the skull then goes into the brain tissue. Depending on the HbO and HbR concentration level the absorption changes. This changes indicate brain activity Ferrari *et al.* [6]. fNIRS is one of the most valuable optical imaging technique due to it's portability, cost effectiveness and ability to use on various population. fNIRS is more flexible that researchers can study brain activity during natrulastic task such as walking, hearing speech Cui *et al.* [2]. Recently fNIRS is advancing to HD-fNIRS which has improved spatial resolution.

2.1.4 Near Infrared Light

Near Infrared Light (NIR) is electromagnetic radiation which length of 650 to 2500 nm. It has larger wavelengths than the visible light. Human eye can detect Near Infrared Light but the wavelength has to be at most 780 nm. NIR light is so useful in medical science because it can penetrate through skin, bone, brain to a considerable depth. This is because of the absorption rate of NIR light by the other biological chromophores in this wavelength. This makes NIR light highly suitable for non-invasive imaging techniques like Near-Infrared Spectroscopy (NIRS) and Diffuse Optical Tomography (DOT) Hoshi *et al.*[8]. Near Infrared Light is absorbed by Oxygenated Hemoglobin (HbO) and Deoxygenated Hemoglobin (HbR) which is why this light is extremely useful in medical neuroimaging. Different wavelength can distinguish HbO and HbR changes in the blood flow which helps to detect brain activity by optical imaging technique Ferrari *et al.*[6]. fNIRS use NIR light to measure concentration of Hbo and HbR which gives information of brain activity Villringer *et al.*[14]. Near Infrared Light is non-invasive that means to use it there is no surgery required or ionizing radiation. Which makes it safer for frequent use and specially in population like infants and elderly.

2.1.5 Functional Magnetic Resonance Imaging (fMRI)

Functional Magnetic Resonance Imaging in another non-invasive optical neuroimaging technique. Which uses the changes of some specific substances in the blood flow to detect brain activity. When a specific region in the brain gets activated the consumption of oxygen increases. This results in a detectable change in the magnetic properties of blood, which fMRI can capture Ogawa *et al.*[12]. fMRI uses different technique than fNIRS. It uses magnetic field and

radio wave to create brain image and activity. The quality of fMRI image is excellent, high resolution which makes it a good tool for mapping functional areas for the brain such as auditory cortices, visual cortex and motor. fMRI can assess the entire brain, including deep structures such as the thalamus and hippocampus. Thalamus and hippocampus are the deepest region of the brain.

2.1.6 fNIRS Source and Detectors

fNIRS sources and detectors are the most important component to measure brain activity by detecting changes in blood oxygenation. fNIRS output of optical image quality highly depends on the placement of sources and detectors on the scalp. Sources placed on the scalp and emit Near Infrared light with range of 650 nm to 850 nm. This light penetrate through the scalp and skull into the brain tissues. Hemoglobin in the blood responsible for absorbing this light. Sources generally use light-emitting diodes (LEDs) or laser diodes. They are used due to their compact size, low cost and ability to emit light at different wavelength which is important to distinguish HbO and HbR concentration in the blood. Another reason is it can emit more focused light which helps to penetrate deeper into the tissue. On the other hand, detectors receive the Near Infrared light that travel through scalp, skull and brain tissue. When NIR light passes this tissues some sort of absorption and scatter happens due to some specific substances of blood. The detectors measure intensity that comes to the detector. It allows to calculate the concentration of HbO and HbR based on how much light is absorbed.

2.2 Related Work

Since this is a novel research problem, direct previous related work is very limited. This chapter is all about past researches and papers related to auditory cortex studies particularly fNIRS studies and other neuroimaging methods. My aim is to find the most relevant work that aligns with my own research. This provides a foundation and context for my study. By reviewing key research papers I can establish the current understanding auditory cortex processing. This review highlights where my research can contribute, particularly the optimal position for source detector configuration.

In [17] they work with the auditory cortex activation .Mainly the focus on the duration of the sound effect. They use fNIRS(Functional Near Infrared Spectroscopy) for their study. The goal was to analyze the amplitude and non-linearity of cortical activation using different duration of sound(8s, 10s, 15s, 20s) and determine the most efficient duration for fNIRS study in future. They mainly use fNIRS to measure the concentration of HbO and HbR levels in response to white noise across different block duration. Total of seventeen adult take part in this study as subject. They find that auditory cortex activation increases with longer stimulation duration. However, it reaches plateau stage after 15 seconds of stimulation.

[9] discuss about the activation in auditory cortex increases linearly as the speech becomes more clearer.The clearer the speech is, the more significant the activation in the auditory cortex, particularly in response to intelligible speech. Intelligible refers when a person talks the spoken language is clear and understandable to the listener. This happens when the speech is easily process by the auditory cortex of the listener and listener does not require extra effort to listen. This study also observe that left auditory cortex gets activated when a person is in effortful listening state or decision making state. This study says that the left auditory cortex plays a crucial role not only in processing clear speech but also in cognitively demanding listening tasks.

In the research of [15] shows us the auditory cortex activate differently when the sound type is different. Researchers use eighteen healthy people as subject for this research with no hearing problem and no subject had history of neurological disorder. They use six distinct type of sound. Those are English, non-English, annoying sounds(such as alarm or unpleasent sound), nature sounds(like rain, wind), music and gunshot sounds. The way they conduct this research is, they use 10s of each sound type on a single subject. Based on this they measure HbO and HbR concentration change in auditory cortex reflecting brain activation pattern in response to these sounds. This study aims to understand how different sound categories are processed in the brain by comparing the changes in HbO and HbR in auditory cortex. Lastly the researchers want to classify each type of sounds and for that they use LSTM(Long Short Term Memory) neural network which is a time seres algorithm. This neural network is commonly used to predict sequence. The concentration change of HbO and HbR in the blood of auditory cortex due to the sound, the LSTM is able classify all the sound types with high accuracy.

In [3] study shows that auditory cortex activates more when the environment is noisy. For this research the researchers test auditory cortex in two different way using fNIRS to measure changes in oxygenated (HbO) and deoxygenated hemoglobin (HbR) concentrations. First, when the environment is noisy participents listent to human voice and researchers measure the HbO and HbR in the auditory cortex using fNIRS. After that, they conduct the study in a quiet environment where they use human voice and measure HbO and HbR in auditory cortex. The result shows that auditory cortex become more active when the subjects are exposed to noise compared to when they were in quiet environment. The findings are background noise enhance brain's effort to process human speech because it needs more cognitive resources to differentiate human speech from environment noise.

Until now I focus primarily on the activation of auditory cortex and discuss it's mechanism and related process in detailed. Now, I introduce couple of other key papers those are highly relevant to my research. These studies provide crucial points that I have integrated into my research work.

[4], this particular paper is one for the important key paper of my research work. NIRFAST, an image modeling tool, forms the foundation for all the simulation work I conduct. Diffuse Optical Tomography or fNIRS is recently widely investigated because of it's non-invasive technique. This is a technique

where near-infrared light (650nm-900nm) injected into the tissue and it measures scattered light that comes back to the detectors thus it can reconstruct images using transport-model-based algorithm. NIRFAST is a MATLAB based software which develop for accurate light propagation in biological tissue. NIRFAST supports both single-wavelength and multi-wavelength imaging, allowing for chromophore concentration and scattering spectra reconstruction. NIRFAST is for both forward model and inverse problem in optical tomography. Forward model is diffusion approximation based on the light transport. On the other hand inverse problem is the reconstruction of images from boundary measurements using least-squares-based techniques. NIRFAST supports both single wavelength and multi wavelength imaging. NIRFAST, allows for complex modeling for both 2D and 3D biological tissue. 2D models are much simpler than the 3D models. 3D models are more representative of clinical settings. In brain imaging NIRFAST can detect the functional changes by calculating the oxygen level and hemoglobin concentrations.

Now, I talk about comparison between two techniques of brain imaging. [5] talks about the comparison between fMRI and fNIRS. fMRI technique serves a very standard service in the field of functional brain mapping due to it's high spatial resolution. However, fMRI has some limitations such as immobility, high cost and the last one is use of metallic implants with subjects. This limitations motivated to look for other modalities for imaging. One of them is Diffuse Optical Tomography (DOT). High Density Diffuse Optical Tomography is more advanced than DOT. [5] focuses on the voxel wise comparison between fMRI and HD-DOT focusing ability to map cortical activity. HD-DOT is an optical neuroimaging technique where use Near Infrared Light to measure the changes of HbO and HbR in particular brain tissues. It is non-invasive, portable and cost-effective compared to fMRI. In this study, researchers do a voxel wise comparison. Then they find that HD-DOT has average localization error of only 4.4mm compared to fMRI. It shows that HD-DOT gives high accuracy in such imaging technique.

CHAPTER 3

Methodology

In this chapter, I will discuss the background of the study, outlining the approach I used for this research as well as the techniques and methods involved.

3.1 Legal, Social, Ethical and Professional Issues

For this project, MATLAB was provided free of charge by the university under an academic license. The use of NIRFAST has been approved by my supervisor, ensuring compliance with the software's licensing terms. Therefore, the work adheres to all relevant software licensing agreements.
This project is based on a simulated head model rather than an original head or patient data. However, it is ethically important to ensure that the simulation is as accurate and representative of real-world conditions as possible. Wrong representation can lead to false assumption.
As this research fully based on simulation, it is essential to present the actual result rather overstating.
In future is this research progress to the real world testing using fNIRS sources, detectors and straps on actual subjects ethical issues such informed consent, privacy and potential risk.

3.2 Environment Setup

My main part is to find the optimal position on the scalp of sources and detectors so that I get maximum result from the auditory cortex when it stimulates by sound effect. For that I was given a 'headmesh' figure 3.1 where all the partition were created before. I have used this given 'headmesh' as my human head and run my simulation using it in Matlab. There are 5 parts in a human head and

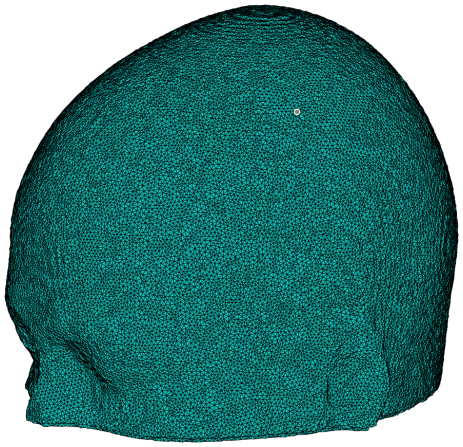


Figure 3.1: headmesh

those are skin, bone, grey matter, white matter and csf. Each part is different to each other when it comes to propagating NIRS light. That's the reason each part has different absorption and scattering coefficient. In table 3.1 shows that

Head Parts	**Absorption**	**Scattering**
CSF	0.0040	0.3
White matter	0.0208	1.0107
Grey matter	0.0192	0.6726
Bone	0.0139	0.84
Skin	0.0190	o.64

Table 3.1: Different absorption and scattering values of each part of head.

each part of head would react differently when light being injected to the head.

3.2.1 Inside the Simulated Head

Figure 3.2 shows the inner structure of the simulated head. This simulation effectively recreates a realistic environment, closely replicating the internal appearance of an actual human head.

3.3 Creating Activation

In this section, I will explain how the activation, or "blob" was created inside the simulated brain. The human brain has two auditory cortices, located about 15 mm above each ear and 10 mm inwards from the scalp. Each auditory cortex is 20 mm in width and 10 mm in height. The left auditory cortex processes human speech, while the right one is responsible for music. Since my research focused only on human voice, I worked specifically with the left auditory cortex. When the human brain detects a voice, the blood flow in certain areas changes, specifically affecting the concentration of oxygenated (HbO) and deoxygenated



Figure 3.2: Inside the simulated head

hemoglobin (HbR). During this process, the absorption and scattering coefficients also change. While this happens naturally in real life, I have recreated this scenario in my simulated head using NIRFAST. Figure 3.3 is the zoomed in

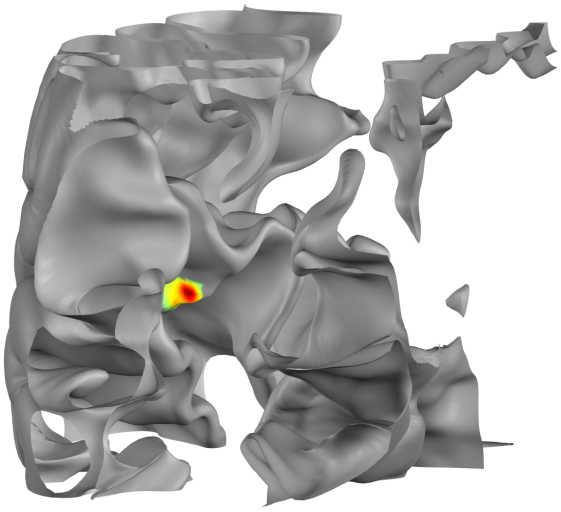


Figure 3.3: Activation Inside the Head

version of figure 3.2. In this closer look, we can see a blob with a different color, representing the activated left auditory cortex, as if the brain is detecting a human voice. This activated area is located exactly 15 mm above the ear and 10 mm inward in my simulated head. Additionally, the absorption and scattering coefficients of this specific area have been modified. All of this done manually. Firstly, I estimated the coordinates of this specific area on the simulated scalp. Then wrote code to create activation inside the simulated head.

3.4 Initial Approaches with Fixed Sources and Detectors

This section covers the initial setup of sources and detectors on the simulated headmesh scalp. These sources and detectors represent the fNIRS light source and detector. I replicated this setup in my headmesh using NIRFAST. Initially, I worked with a fixed position for the sources and detectors, and the number of optodes was kept limited. At first I try with the fomation showing in figure

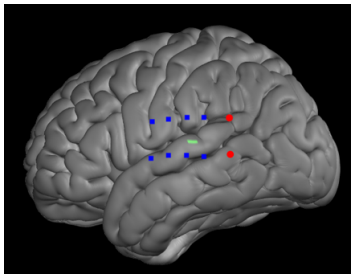


Figure 3.4: Formation 1

3.4. Here red optodes represent the sources and the blue optodes are detectors. In this scenario I use two sources and eight detectors. Here each source creats a channel with every detectors. So, one source creates eight channels. All together this setup results in sixteen channels. In my second attempt I use 4 sources and

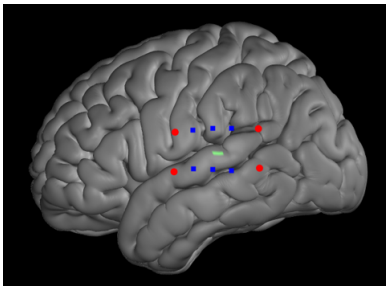


Figure 3.5: Formation 2

second detector as shows in figure 3.5. In this formation the total number of channels is twenty four.

3.4.1 Problem with fixed and limited optodes

There could be some problem when fixing the optodes positions for sources and detectors. For example, I might miss valuable source-detector (SD) pairs that could provide better data, but aren't selected as sources and detectors. Another issue is the potential loss of important channels in areas where no sources or detectors are placed, as the number of optodes is quite limited in the formation mentioned above.

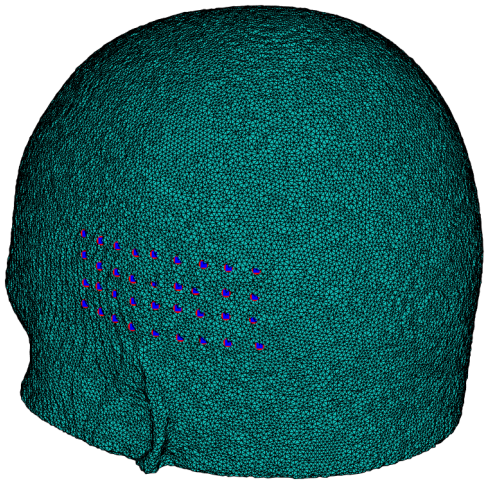


Figure 3.6: Final Formation

3.5 Maximizing Source-Detector Combination

In this section, I explain how I solve the issue that arises when the source-detector (SD) pairs are fixed and limited. To overcome this, I positioned a greater number of optodes in the area of the auditory cortex. Specifically, I used thirty-six optodes to ensure they cover a larger area, minimizing the risk of missing any important information. I create this optode manually. The way is I look at my mesh and find these x,y,z coordinates. In table 3.2 shows the coordinates of thirty six coordinates. In this case, I do not fix any optodes as source or detector. Instead, I use all optodes as both sources and detector. So, in this way a large number of channel is being created. This approach allows a coverage of big area around the auditory cortex. So that, If the left auditory cortex has some activation, the channels can capture this activation. In figure 3.6 shows all the 36 optodes. In this setup each optodes is represented as both blue and red color. This dual role of each optode help more large number of data collection. Moreover from figure 3.6 it is clear that a large area is covered just above 15 mm from the ear of this simulated head mesh so that I do not miss any information if there any activation happens in the simulated auditory cortex.

Another point is the dramatic increase in the number of channels. By this formation, I can create 1260 channel which is huge compared to previously discussed formation 1 and formation 2. This significant increase in channel number means more detailed and precise information can be captured from the auditory cortex, reducing the risk of missing important information. This large number of channels increases the reliability of the simulation result.

This formation represents a major improvement in terms of both spatial coverage and data collection capacity, making it a more effective approach for studying auditory cortex activation.

Optode Number	X position	Y position	Z position
1	-75.497	-60.6597	23.3938
2	-76.527	-59.7415	1.8209
3	-79.543	-50.3532	2.309
4	-78.713	-49.7402	23.3223
5	-82.487	-40.9129	1.9661
6	-83.493	-30.3324	1.1809
7	-82.615	-20.173	1.3609
8	-81.516	-10.394	1.5999
9	-79.482	-1.479	0.1183
10	-81.556	-39.6442	23.9549
11	-82.46	-30.7959	22.5415
12	-81.458	-20.068	22.293
13	-80.274	-9.984	21.2728
14	-77.486	0.937	20.5718
15	-79.49	-50.6068	12.0683
16	-81.452	-40.0453	11.1484
17	-83.534	-29.9768	10.9893
18	-82.522	-20.54	10.9675
19	-81.297	-10.417	10.5064
20	-78.752	-0.313	11.8946
21	-76.728	-57.6585	11.356
22	-72.467	-70.0335	2.1359
23	-67.265	-80.3862	1.9196
24	-72.416	-70.0897	12.3359
25	-66.53	-80.3672	11.6999
27	-71.202	-70.3243	23.2168
28	-65.281	-80.2498	22.9951
29	-78.471	-0.145	-9.789
30	-80.723	-10.806	-9.7021
31	-82.44	-19.627	-9.697
32	-83.537	-29.6207	-10.1978
33	-81.784	-40.6013	-10.2447
34	-78.927	-51.2362	-10.4246
35	-75.544	-60.8856	-10.0527
36	-72.445	-71.0607	-9.3775

Table 3.2: Coordinates of 36 optodes.

3.6 Finding the Important Channels

In the final formation I have 1260 channels. All of these channels are not equally important. My objective is to find those channels which carry most important information from auditory cortex. In the practical scenario those are the important channel which are having significant amount of absorption or scattering. Because, when a stimulation is going on in the auditory cortex the concentration level of HbO and HbR changes in the blood flow.

If near-infrared light propagates into this specific brain region during the time of activation of stimulation, absorption or scattering may occur due to the altered HbO and HbR levels. Since I am simulate this scenario, I need to calculate the extent of absorption or scattering for each channel to determine their significance.

To find out the most significant channels I need to calculate Jacobian Matrix also refers as 'sensitivity matrix'. It calculates how sensitive a channel is due to the change of optical properties in the 'blob' or stimulated area. This matrix helps me identify which channels are most responsive to the hemodynamic changes in the auditory cortex. In other words, it pinpoints the channels that carry the most valuable information about brain activity.

3.6.1 Light Propagation in the Head

When Near Infrared light shine into the head it propagates through various layers such scalp, skull, grey matter, white matter etc. As this light passes through this tissues, the light is absorbed or scattered. The level of absorption or scattering changes where blood flow and oxygenation level increases in a specific region. Typically the increment is based on the brain activity. If one part of the brain is active the level of blood flow and oxygenation level increases.

3.6.2 Jacobian Matrix

Jacobian matrix also known as sensitivity matrix plays a very important role in a technique like fNIRS Zimeo *et al.*[18] and Diffuse optical tomography. Jacobian matrix calculates how sensitive each channel is to the changes in the optical properties such as absorption and scattering at every point in the simulated headmesh.
Each elements in this matrix shows how a small change in absorption or scattering at a particular location in the simulated brain affects the intensity of light detected by a particular source detector pair. In other words, it tells us a numerical measure how much light detected by a channel in response to change in optical properties in a particular region of the brain. These properties are absorption and scattering properties of the brain tissues.
For instance, if there is any stimulation going on in the auditory cortex Jacobian matrix shows how much absorption or scattering occurs for a particular light source or channel by auditory cortex tissues. A large value in a row of the Jacobian matrix says, that particular channel is highly sensitive to the change of brain tissue. On the other hand, rows having small values indicates those channels are less sensitive. On other words those are the channel carry less

information from that particular brain region.
This nature of Jacobian matrix helpsto capture complex interactions of light as it propagates through the heterogeneous tissues of the brain. By analyzing Jacobian matrix it is easy to find out the most informative channel for detecting changes in brain tissue due to detecting sounds, visual effect etc.

3.6.3 Jacobian Relates to Absorption and Scattering

Jacobian captures how light interact with different parts of the brain. If jacobian tells that a channel is significantly sensitive it means that the changes in absorption (due to blood flow changes from brain activation) or scattering in that region will significantly affect the light measured at the detectors. On the other hand, if the the jacobian matrix says that low sensitivity that means absorption and scattering have little effect on the measured signal. This indicates that this specific channel is not carry any information due to the auditory cortex activation.

CHAPTER 4

Results

In my study there are total 36 optodes and these optodes are configured as both sources and detectors at the same resulting in 1260 channels. To measure each channels capability I create an activation in the simulated headmesh which is mimicking the response of auditory cortex when human brain detects sound. Based on the activation created in the simulated headmesh, I calculated the jacobian matrix. This jacobian matrix show the sensitivity of each channel to changes in the optical properties in the headmesh.

After getting the jacobian matrix I then I perform normalization on the jacobian matrix so that, each value range becomes 0 to 1. 1 being the highest sensitivity and 0 refers no sensitivity at all.

Channel Number	**Source Coordinate**	**Detector Coordinate**	**Sensitivity**
428	[-78.47, -0.14, -9.79]	[-77.49, 0.94, 20.57]	0.5695
796	[-65.28, -80.25, 23.00]	[-67.27, -80.39, 1.92]	0.9851
860	[-78.75, -0.31, 11.89]	[-66.53, -80.37, 11.70]	0.5101
874	[-72.44, -71.06, -9.38]	[-66.53, -80.37, 11.70]	0.5085
875	[-65.96, -82.51, -9.64]	[-66.53, -80.37, 11.70]	0.7769
933	[-67.27, -80.39, 1.92]	[-65.28, -80.25, 23.00]	1.0000
945	[-65.96, -82.51, -9.64]	[-65.28, -80.25, 23.00]	0.9054
959	[-77.49, 0.94, 20.57]	[-78.47, -0.14, -9.79]	0.6221
1250	[-66.53, -80.37, 11.70]	[-65.96, -82.51, -9.64]	0.7816
1252	[-65.28, -80.25, 23.00]	[-65.96, -82.51, -9.64]	0.9108

Table 4.1: Most Sensitive Channels

Table 4.1 shows the top sensitive channel among 1260 channels which are having most of the information. There 1260 channel however, every channel is not important. So, it was crucial to filter out the less significant ones to focus on those that genuinely reflect the neural activity of interest.

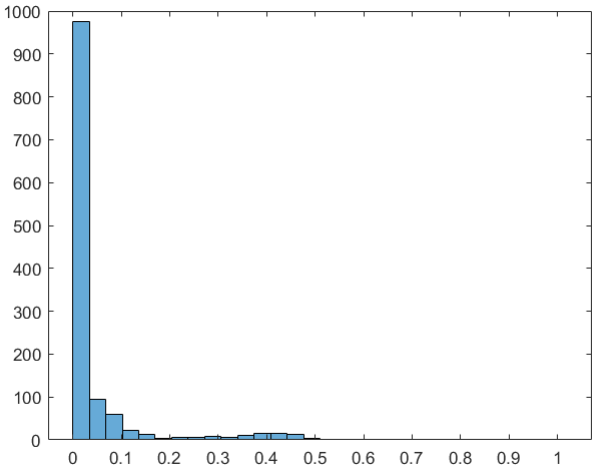


Figure 4.1: Sensitivity Histogram

So, I check the histogram of sensitivity. 4.1 indicates that if I keep sensitivity threshold 0.2 I get almost every source detector which are important. I don't want to take any risk as I can see some small peaks at sensitivity 0.3 to 0.5. That is the reason I set sensitivity threshold 0.5 and after all I get only 10 important channels out of 1260 channels which are having most of the information about the activation or 'blob' I created in my simulated headmesh. Based

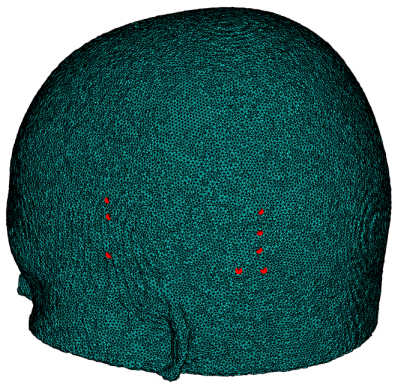


Figure 4.2: Important Sources

on the channel I get from jacobian matrix I find out that from which source

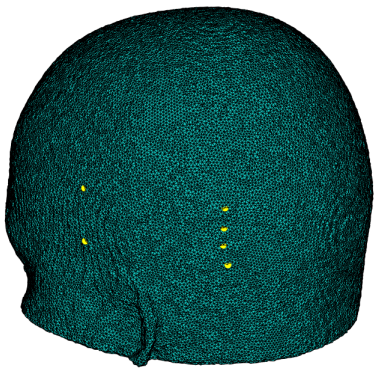


Figure 4.3: Important Detectors

detector pairs this channel is created. Then, I find out the important sources from 36 optodes. Figure 4.2 shows the important sources and figure 4.3 shows the important detectors.

Results shows that 10 out of 1260 met my sensitivity threshold meaning that they have the most significant information about the 'blob' I created in the headmesh. These channels are crucial to find out the actual changes in the specific region where activation occurs. By just selecting this channel and optodes placements I can optimize fNIRS measurement process and improve the data quality. Next step is to find out the source detector pairs of these ten most important and sensitive channels. While using the Jacobian matrix, I pinpointed the source-detector pairs that form these key channels.

In summary, by the use of jacobian matrix and chooing a threshold I am able to filter 1260 channel to only 10 channel most important channels. This channels ensure me the collected data from this channel is the perfect representation of the activity that occurs in the simulated auditory cortex in the headmesh.

CHAPTER 5

Discussion and Future Work

5.1 Discussion

The main goal of my project is to simulate the light propagation inside the head where auditory cortex is and find out the most sensitive channels. It allows me to select the best and optimal position of the sources and detectors positioned on the scalp. I am using Near Infrared Spectroscopy(NIRS) to find out how Near Infrared light reacts when it interact with activated brain tissues specifically auditory cortex region. The Jacobian matrix indicates that which source detector(SD) pairs responds better or having important insights about the change in the auditory cortex. Here important insights refers the how much absorption or scattering occurs for each channel due to the change of HbO and HbR in the auditory cortex. By doing this I can find out the most important pairs positioned on the scalp.

The results from this research shows that there are some certain SD pairs hold higher sensitivity in other words these pairs are the most effective during any changes in auditory cortex. The findings of this research is very important as in practical application it can enhance the accuracy and reliability of auditory cortex activity monitoring. The most important part of the research is the finding the best position for the optodes on scalp for fNIRS technique leading better detection of any activation that occurs in the auditory cortex.

For this research work, I have used simulated head model which helped me to take control over the experiment variables and also gives me chance to test various configurations of the optodes and other parameters. The activation or the 'blob' in the specific region of the brain showed that the current configuration could detect any changes in the optical properties of auditory cortex. This activation is acting as the actual activation of auditory cortex that occurs in the practical situation. However, a very important note is that this is a simulated work which means the results from this research are only approximation of real

world situation. Simulation can not catch all the complexities that occurs in the tissues of human body including individual anatomical changes in the tissues.

Reconstruction of image due to some optical changes in this system is effective as well as it is subject to potential errors. And this inverse problem in optical tomography is ill-posed. That means small error could lead to significant error in reconstructing image. By using Tikhonov regularization we can mitigate this problem however there still could be some challenge mapping optical signals.

Another issue is the selection of the region of activation and assumption of this regions properties (absorption and scattering) directly effecting the results of this research. But this head model does not fully represent the human brain anatomy and real world application may need customization for each participants. for future studies I aim to incorporate more individualized and anatomically accurate models to improve the fidelity of the simulations.

The findings of this study highlights the importance of optimal position of the sources and detectors on the scalp in fNIRS based technique. By selecting effective channels fNIRS studies can give more accurate results of human auditory cortices. This approach is particularly valuable in clinical and cognitive neuroscience research, where precise localization of neural activity is crucial.

5.2 Future Work

This study show some important information about the most sensitive channels and placement of auditory cortex on to the scalp. Still there are some work to be done in the future.

5.2.1 Validation the Simulated Result

The very next step is to validate the simulated result on a real world situation. For this I need to implement a physical head strap where all the sources and detector are placed as I have my final formation shown in figure 4.2 and 4.3. This validation is critical to ensure that the findings from the study is consistent with real world fNIRS measurements. This validation step will help us to understand the discrepancies between natural world and simulation also the nature of the simulation.

5.2.2 Incorporating Anatomical Variability

Another work I aim for is, incorporating anatomical variability in the simulated head model. As human head vary in size, shape and tissue composition. So, employing MRI based could enhance the sensitivity accuracy for individual participants. This kind of change would need unique geometry for each participants and personalised optode placement.

5.2.3 Optical Property Mapping

The refinement of tissue should be done in future. I have changed the optical properties of tissues in the auditory cortex of the simulated brain. The prop-

erties are absorption and scattering however, the changing of this properties depends on the age, health status and brain region. This changes will provide more realistic representation of how light travel through the scalp to the brain regions. Morever, lead us to better sensitivity analysis and better optode placement.

5.2.4 Testing on Different Wavelength

My current research focused on only one wavelength for propagating the light into the brain cell. While fNIRS allows to study on multi wavelength. I can use other wavelengths to check how light propagates into the brain and capture informative information.

In conclusion, this project marks an important first step in exploring how optical imaging can be used to detect brain activity through simulations. The findings are promising, offering some valuable insights, but there is still a long way to go before we can develop truly reliable, non-invasive brain imaging tools. This work sets the stage for future research, providing a foundation to build on and refine as we strive to improve our understanding of brain function.

CHAPTER 6

Conclusion

This project has been an important step in understanding the placement of optodes for optical imaging that can be used to detect brain activity specifically auditory cortex activity through simulations. The use of tools like NIRFAST and MATLAB has allowed us to explore the sensitivity of different source-detector pairs and identify which channels carry the most useful information. By focusing on specific areas like the auditory cortex, I have been able to simulate activation and study how light interacts with the brain's different region.

Throughout this work, I developed a model that helps us visualize and assess the sensitivity of various source detector channels. This is crucial because knowing which channels are the most sensitive is the key to gathering accurate information about auditory cortex activity. Although this is still a simulation-based study, it gives us valuable insights into how we might use this technology in real-world applications such as finding the best placement for the sources and detectors for fNIRS experiments.

CHAPTER 7

Appendices

7.1 Appendix 1

My full code is uploaded at https://git.cs.bham.ac.uk/projects-2023-24/mxh1252. There are several files. To get the access of the important part we have to go headmesh->demofullhead.

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